# Sainagesh Veeravalli

(470)-807-5170 nageshv.9832@gmail.com linkedin.com/in/nagesh9832 github.com/nageshnen

## EDUCATION

**THE UNIVERSITY OF GEORGIA | 2023**
*Master's in Computer Science.*

**K.L University**
*Bachelor's in Computer Science Engineering.*

## TECHNICAL SKILLS

**Languages**: Python, Rust, Go, Java.
**ML/DL Frameworks**: TensorFlow, Pytorch, Keras.
**ML Tools**: MLflow, Weights & Biases
**Computer Vision**: OpenCV, FER, YOLO, GAN's.
**Cloud Technologies**: GCP, AWS.
**CI/CD**: Docker, Azure Pipeline, GitLab CI/CD.
**Frameworks**: Django, Flask, Spring, REST api.

## CERTIFICATIONS

Google Cloud Certified Generative AI Leader
ANAPLAN LEVEL-1 MODEL BUILDER
SAS Advanced Programmer.
Automation Anywhere.
CISCO Packet Tracer.
Google Data Analytics Professional.
Google IT Support Certificate.

## PROFESSIONAL EXPERIENCE

**The Home Depot** — **May 2024 - Present**
*Software Engineer 2* — *Atlanta, GA*

* Partnered with business stakeholders to analyze and refine existing processes and payout calculation methodologies, enhancing accuracy and transparency.
* Designed and implemented Tableau dashboards, leveraging data from GCP and integrated with Salesforce CRM, to provide actionable insights and performance tracking for the sales team.
* Automated weekly reporting processes within Anaplan, reducing manual effort by 8 hours per week and improving reporting timeliness.

**Skills**: Python, GCP, Anaplan, Tableau.

**Qverge** — **Mar 2024 - May 2024**
*M.L Engineer* — *Alpharetta, GA*

* Led a team of 5 in spearheading a Video Sentiment Analysis project, employing Deepface and FER models to decipher emotions and sentiments from video content, contributing to enhanced understanding of user engagement and feedback.
* Collaborated on Gas Cylinder Detection, Tracking, and Monitoring project using YOLOV8x model with CVAT for custom data annotation. Employing MLflow for efficient project management.
* Implemented Agile methodology with daily standup meetings and facilitated streamlined collaboration by utilizing Confluence and Azure Cloud for model deployment in project workflows.

**Skills**: MLflow, DeepFace, SSD, AWS, HuggingFace, Flask, Docker.

**Pinnacle Leasing and Management** — **May 2023 - Feb 2024**
*Software Engineer* — *Alpharetta, GA*

* Employed Azure DevOps for ticket management, ensuring efficient issue tracking, and maintained the latest code base with Azure Repo for code version control and collaboration.
* Implemented Azure Pipelines for continuous integration and delivery, enhancing development and deployment. Utilized AWS services like S3 and CloudFront for high-performance website hosting.
* Automated infrastructure provisioning with Terraform across various web environments for enhanced resource management and scalability, while also creating user-friendly documentation sites with MkDocs.

**Skills**: Terraform, AzureDevops, AWS, SonarQube, Figma, Mkdocs, Postman, ZohoCRM,.

## PROJECTS

**Gated Recurrent Unit** **GitHub Repo**

*python/scala*

- GRU model consists of input, hidden, and output layers. It uses a set of hyperparameters like batch size, maximum number of epochs, and optimization techniques like RMSProp or Adam for training.
- Implemented the forward and backward pass of the GRU model. The forward pass computes predictions based on input sequences, while the backward pass calculates gradients for weight updates during training.
- Utilized a sample lake level dataset to perform testing and validation of the GRU model. This dataset is used to evaluate the model's performance in forecasting lake levels, demonstrating its practical application.

**3-D Object Detection and Identification** **GitHub Repo**

*Python*

- Seamlessly aligned high-resolution LiDAR point clouds with camera images from the KITTI dataset, establishing tightly synchronized data pairs crucial for advanced object detection models and cross-modal insights.
- Leveraged Frustum PointNet and Frustum CloudNet to extract rich features and enhance object detection accuracy within the defined regions of interest (frustums).
- Integrated frustum extraction, feature extraction, and visualization to develop a robust system for 3D object detection in complex real-world environments.

**Image Search Engine** **GitHub Repo**

*python*

- Developed a CBIR image search engine, integrating feature extraction and similarity measurement techniques.
- Elevated search precision through pioneering a background-based image retrieval system, utilizing advanced feature-based similarity calculations, including color histograms and texture descriptors
- Evaluated and fine-tuned the system with diverse datasets, showcasing its practical effectiveness.